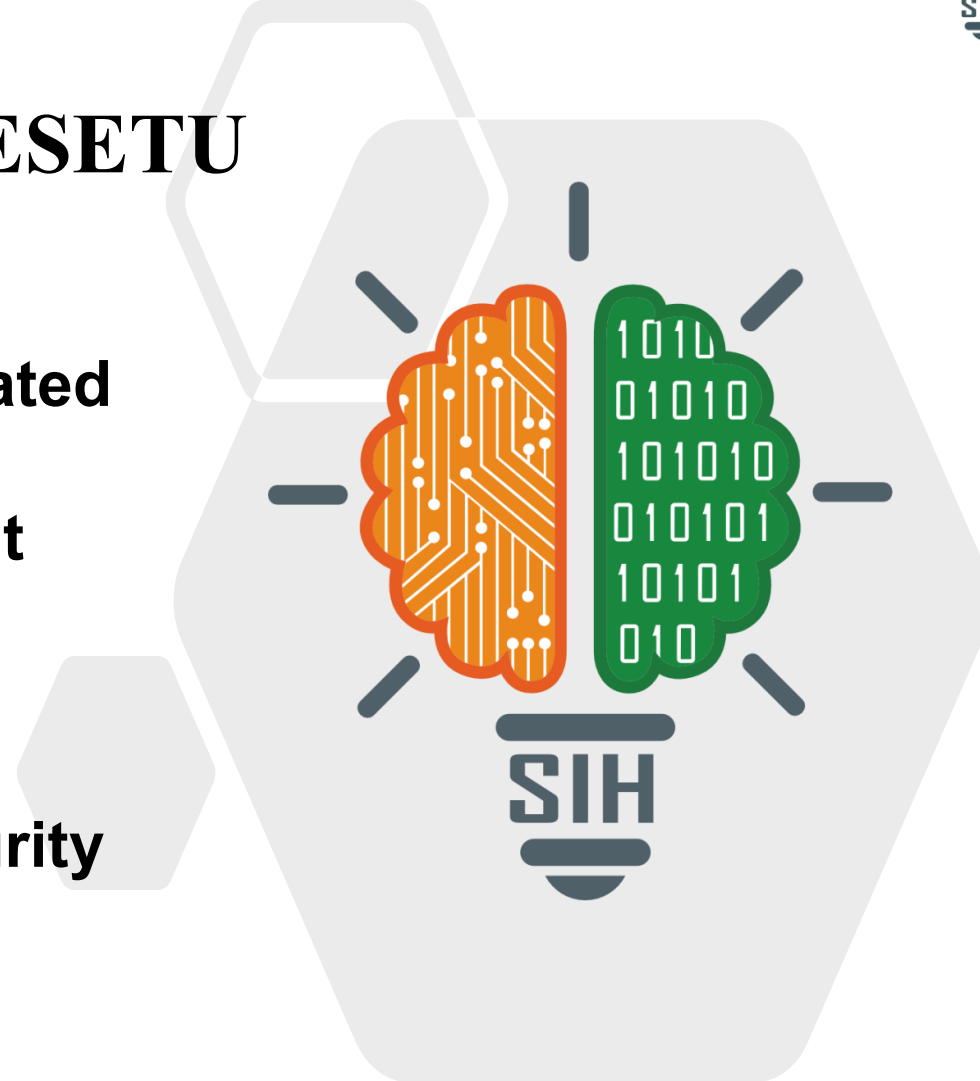


SMART INDIA HACKATHON 2026



TRACESETU

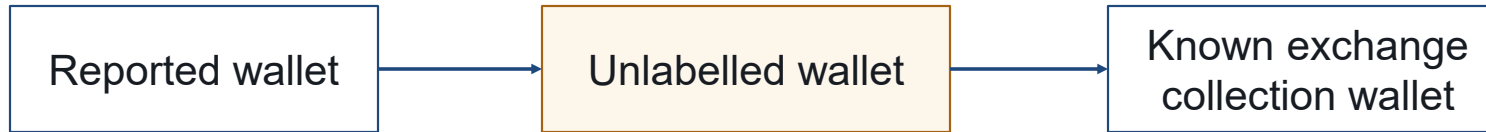
- **Problem Statement ID – PS26182**
- **Problem Statement Title – Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs**
- **Theme – Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID – 170835**
- **Team Name – ANANTHA**



THE INVESTIGATOR'S PROBLEM

Officers often start with a wallet address, not a service name.
Funds may cross several wallets before reaching an exchange.
They need to know which service to approach with a lawful request.

THE ADDRESS WE NEED MAY BE ONE STEP EARLIER



Is the middle wallet already an exchange deposit address?

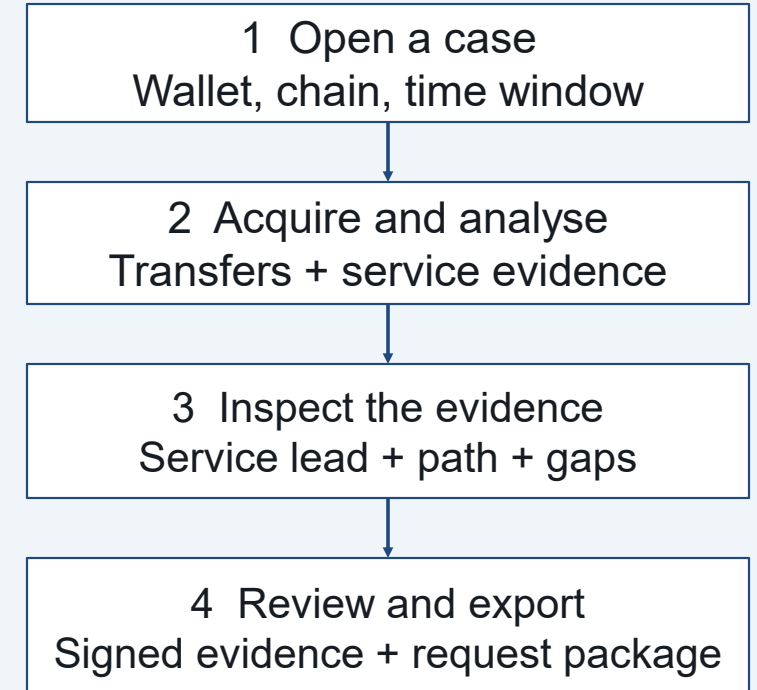
PROPOSED SOLUTION

Trace transfers and check public/provider service labels.
Test possible deposit wallets, then return the first supported custodian on each path, with the evidence and missing links.

OUR FOCUS

Explain the recipient. Challenge the evidence. Choose the next useful check.

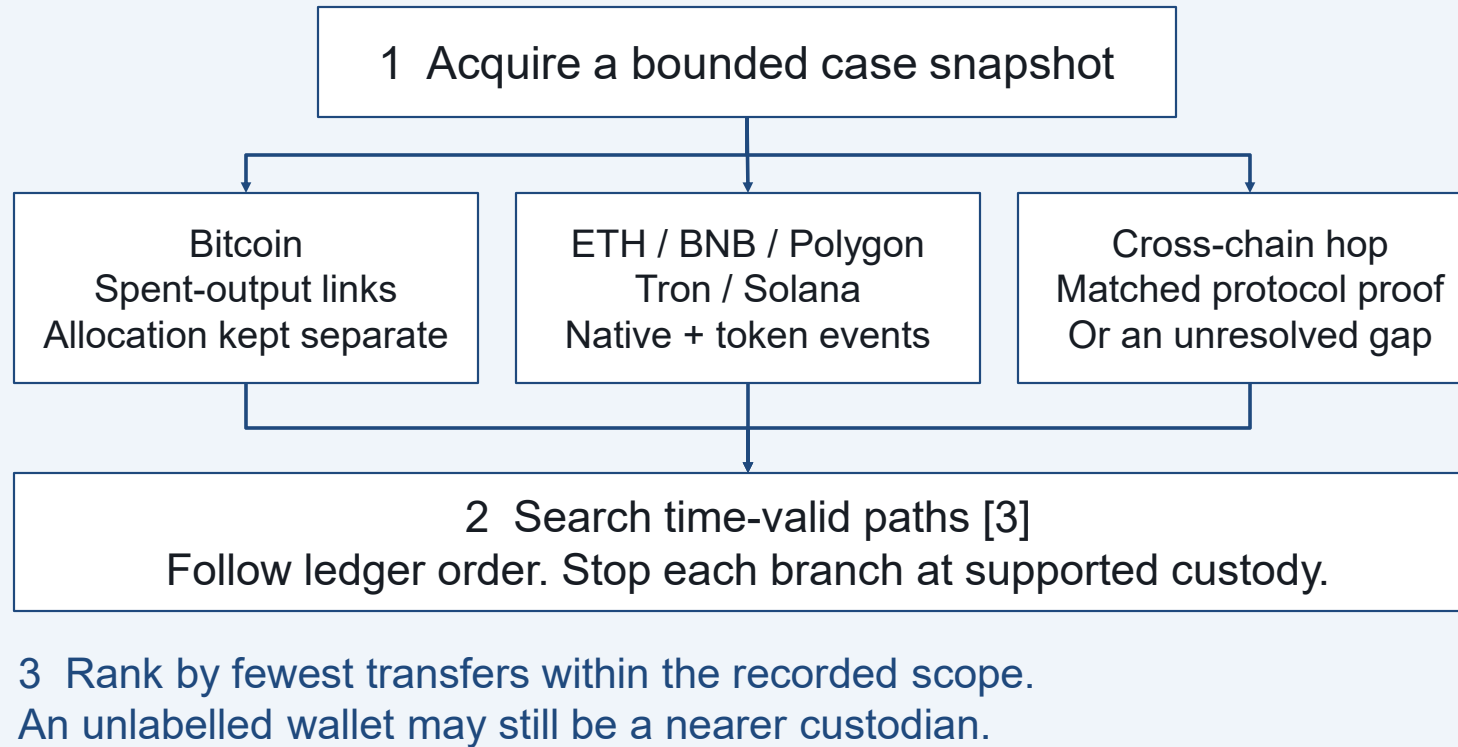
OFFICER WORKFLOW



SAHYOG access is pending.
Current exports remain NOT_SENT.

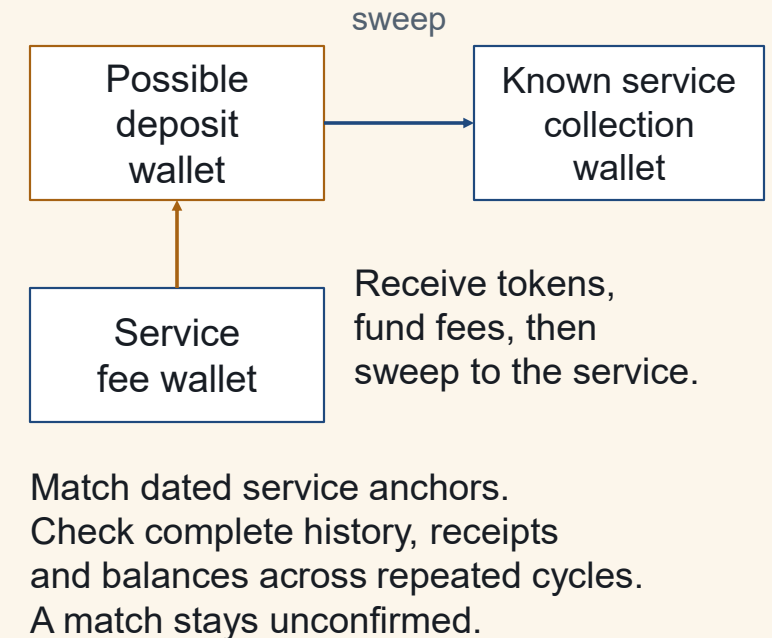
VASP = exchange or crypto custodian

CHAIN-AWARE TRACING



DEPOSIT CANDIDATE [1, 2]

Separate research module



TECH STACK: EACH TOOL HAS A SPECIFIC JOB

FastAPI / Python: APIs + graph rules • React / TypeScript / Cytoscape: investigator review + graph view
PostgreSQL + workers: queued jobs • SQLite: local demo • SHA-256 / Ed25519: export integrity

FEASIBILITY: A 36-HOUR DELIVERY PLAN

Scoped sprint using our existing code foundation

0–6 h	Scope case + acquire Bitcoin history
6–16 h	Trace + stop at sourced custody
16–24 h	Show gaps + challenge a source
24–30 h	Cap queries + review/export evidence
30–36 h	Test failures + rehearse a live-data case

EVIDENCE THAT THE CORE IS BUILDABLE

Public Bitcoin check: 2 transactions, 3 outputs.
Transfers reproduced; no VASP established.

Research module: 64 synthetic checks pass.
Integration and independent live accuracy are pending.

Six adapters exist; six-chain field validation is staged.

PRACTICALITY: MINISTRY DEPLOYMENT

Foundation already built

Separate API/workers, PostgreSQL job claims,
case permissions, audit history and signed exports.

Scale-up design

Bound each trace; cache data; add worker capacity.
Shared quotas and agency fairness still need work.

Load tests, SSO/MFA and SAHYOG access: pending.

REAL-WORLD BOTTLENECKS & CONTROLS

Free APIs have limits → budget calls and show gaps.
National coverage needs governed data access.

Labels may be wrong → retain sources; review claims.
Patterns alone cannot prove wallet ownership.

Pilot first; measure queue delay, cost and error rates.

NOVELTY: THE DECISION METHOD

Bounded first-custody search

Find first custody per path and expose nearer gaps in an inspectable certificate.

Let missing evidence guide queries

Prioritise unresolved nearer branches within a fixed request budget.

Challenge and recompute the result

Remove a source; retain previous results.
Review signed feedback before relabelling.

CONTROLLED EXAMPLE: AN EARLIER DEPOSIT LEAD

Fictional path: reported wallet → possible deposit → known hot wallet.

Comparison from the separate research experiment, not a live accuracy result.

Evidence available	Label-only search	Candidate method
Complete toy pattern	Known hot wallet 2 transfers away	Possible deposit 1 transfer; unconfirmed
Missing fee evidence	Known hot wallet Still reachable	Unresolved candidate Ask for more evidence
Customer imitates pattern	Cannot prove control	Cannot prove control Keep it unconfirmed

OFFICER OUTCOME: A SUPPORTED RECIPIENT OR A SPECIFIC NEXT CHECK

SUPPORTED SERVICE

Verify recipient; prepare the request.

DEPOSIT CANDIDATE

Seek independent confirmation.

UNRESOLVED

Get more evidence or record the gap.

TEST THE CONTRIBUTION AGAINST LABEL-ONLY TRACING [5]

Known graph methods; our workflow must earn its value in recipient accuracy, coverage, query cost and officer time.

RESEARCH USED IN THE DESIGN

[\[1\] Victor · FC 2020](#)

[\[2\] Brechlin et al. · 2025](#)

[\[3\] Wu et al. · PVLDB 2014](#)

[\[4\] Kappos et al. · 2022](#)

[\[5\] Lubbertsen et al. · 2025](#)

Deposit-address forwarding: a clue requiring known service anchors.

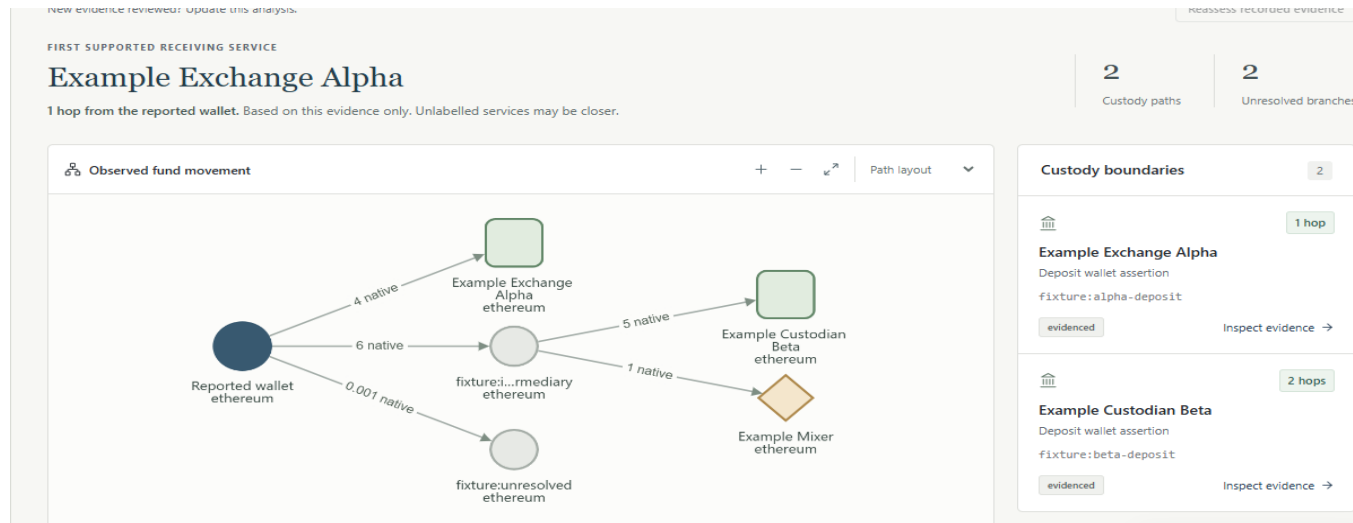
Service fee funding + collection; Evonax confirmed roles using service records.

Temporal graphs: reject transfer paths that run backwards in time.

Bitcoin peel-chain clustering: a guarded future extension.

Evaluate attribution against independent ground truth and address roles.

ACTUAL PROTOTYPE: OBSERVED TRANSFERS, SERVICE EVIDENCE AND NEXT ACTIONS



[Public source code & research](#)

github.com/notvinayaks/TraceSetu

Complete work: product/live-foundation

[Watch the prototype walkthrough](#)

youtube.com/watch?v=V_qB6dL934w

Screenshot: fictional training case in the real app.

Public Bitcoin testing is reported on slide 4.

The video demonstrates the existing prototype.